Comprehensive Overview of Automated Notifications, Communication Pipelines & Policy Guardrails

Math 105L & 106L: Laboratory Calculus I & II | Duke University

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1 System Architecture & Objectives

The instructional operations for Duke University's Laboratory Calculus sequence (Math 105L and 106L) utilize an automated, high-throughput management infrastructure. This architecture integrates the **Gradescope API**, **Canvas Gradebooks**, **Microsoft Graph API** (OAuth2 authenticated via Duke NetID), and cloud-triggered **GitHub Actions** workflows.

The notification subsystem ensures three core operational outcomes:

1. **Student Transparency & Equitable Support:** Immediate feedback on assignment status, procedural extension granting for defective scans, clear visibility into retake opportunities, and regular comprehensive performance audits.
2. **Teaching Assistant (TA) Alignment:** Proactive grading schedule reminders, rubric completion tracking, and lab section coverage coordination.
3. **Instructor Audit Logging & Governance:** Complete auditability of every deadline adjustment, extension grant, threshold alert, and student message dispatched by the autonomous pipelines.

Key Operational Email Routing

- **Instructor Academic & Course Administration:** rb6@duke.edu (all Gradescope updates, student notification summaries, TA schedules, and no-pass threshold alerts).
- **Instructor Public, Research & Civic Infrastructure:** rann@rannbar-on.com (election forecast pipelines, switchover confirmations, public website updates).
- **Automated Sender Identity:** rb6@duke.edu via Microsoft Graph REST API using locally persisted OAuth2 credentials (~/.duke_token_cache.bin).

2 Student Notifications

Student communications are strictly governed by automated validation routines to prevent communication fatigue and enforce equitable course policies.

2.1 Catalog of Student Notification Types

Notification Type	Cadence / Trigger	Audience / Routing	Operational Logic & Actions
Due-Today Reminder	Due date at 8:00 AM	Unsubmitted enrolled students	Alerts students of upcoming 11:59 PM deadline on Gradescope. Limited strictly to *current* assignments.
Scan Defect & Extension	Post-submission verification	Student; Grader CC'd	Triggered when a submitted PDF has missing pages, inverted page order, or photo capture flaws. Automatically grants a **12-hour extension** ($\max(now+12h, due+12h)$) on Gradescope and requests resubmission via mobile scanner apps.
Missing Deadline Alert	Morning after due date	Unsubmitted students	Alerts student that a current assignment deadline passed without a recorded submission.
Missed Quiz & Retake Notice	Post quiz grade release	No-Pass or missing quiz	Informs student that the original quiz score is recorded and the corresponding Quiz Retake is open on Gradescope with dates and submission criteria.
Retake Score Release	Exactly 1 wk + 1 hr post-retake open	Retake participants	Reports graded retake score, pass/no-pass status, and updated mastery progress. Guarded by the **Download Delay Rule**.
Tuesday Resubmission Reminder	Tuesdays at 9:00 AM	Students with eligible prior assignments	Reminds students of weekly Tuesday resubmission deadlines for eligible prior worksheets and lab reports.
Comprehensive Status Report	Mid-semester & milestones	All enrolled students	Complete individual transcript: worksheets passed/NP, quiz scores, retake statuses, lab marks, and overall trajectory.
100% Perfect Score Praise	Immediate upon grade finalization	Student recipient	Congratulatory message sent whenever a student earns a perfect 100% on an individual worksheet or lab report.
Official Reschedule Notice	Calendar adjustment	All enrolled students	Official broadcast when an assignment deadline is universally rescheduled (e.g., Worksheet 10-3 extension to Nov 9).
Video Project Group Allocation	Project kickoff	Assigned student groups	Distributes assigned group teammates, project prompt instructions, benchmark deadlines, and submission instructions.
Solo Lab Report Notice	Upon submission of solo report	Solo submitters	Warns student that lab reports are collaborative group assignments and requests linking lab partners on Gradescope.

2.2 Mobile Scanning Application Requirement

When a worksheet submission fails automated page verification or exhibits illegible camera photography, the automated notification specifically advises the student:

> *"Please submit using a mobile document scanning application rather than loose camera photos or flatbed scans. We strongly recommend the official Gradescope mobile app (iOS/Android), or built-in utilities such as Apple Notes (iOS), Google Drive Scan (Android), Microsoft Lens, Adobe Scan, or Genius Scan. An automated 12-hour extension has been posted to your Gradescope account."*

3 Teaching Assistant (TA) Notifications

Graduate and undergraduate Teaching Assistants play an essential role in grading laboratory reports, leading problem sessions, and proctoring. The automated pipeline coordinates TA workflows without manual administrative overhead.

3.1 Catalog of TA Notification Types

Notification Type	Cadence / Trigger	Audience	Description & Payload
Thursday Grading Reminder	Thursdays at 4:00 PM	All Math 105L/106L TAs; `rb6@duke.edu` CC'd	Comprehensive reminder of assignments closing that evening, assigned grading rubrics, grading deadlines (typically Sunday evening), and rubric guidelines.
Monday Grading Follow-up	Mondays at 9:00 AM	TAs with pending queues	Identifies lingering unfinalized rubrics, incomplete submissions, or unresolved student regrade requests.
Daily Lab & Shift Schedule	Daily at 7:30 AM	Active daily TAs; Instructor	Summary of active lab shifts, room assignments, problem session schedules, and cumulative years of TA experience for peer mentorship.
Prospective TA Recruitment	1 week after initial callout	Past qualified TAs	Follow-up reminder to past successful TAs who have not yet submitted their When2meet / Qualtrics availability for the upcoming semester.

4 Instructor Notifications (Sent to Rann Bar-On)

To ensure absolute oversight over all autonomous operations, every systemic change, student communication, and boundary condition triggers an immediate or batched report to the instructor (`rb6@duke.edu` or `rann@rannbar-on.com`).

4.1 Catalog of Instructor Notification Types

Notification Type	Delivery Cadence	Destination	Content & Operational Triggers
Assignment Creation / Update Audit	Immediate upon API action	`rb6@duke.edu`	Dispatched whenever an assignment is created or modified on Gradescope (title, dates, submission windows, extensions, or rubrics). Must include the explicit reason for the modification.
Daily Morning Student Digest	Daily between 8:00 AM–12:00 PM	`rb6@duke.edu`	Batched audit of every student communication dispatched over the prior 24 hours, parsed from `grades/.sent_student_notifications.json`.
Re-Retake Completion Immediate Alert	Immediate upon submission	`rb6@duke.edu`, CC `rann@rannbar-on.com`	Sent when a student requiring re-retakes finishes an individual assignment or fulfills all needed re-retakes. Includes student name, score, and progress.
Critical No-Pass Threshold Warning	Upon grade calculation	`rb6@duke.edu`	Triggered whenever a student accumulates their **5th No-Pass on worksheets**, **3rd on quizzes**, or **2nd on labs**.
Excessive No-Pass Block Notification	On extension evaluation	`rb6@duke.edu`	Fired when a student exceeds **8 worksheet NPs**, **4 quiz NPs**, or **2 lab NPs**. The pipeline blocks extensions until explicit instructor authorization.
Semester-End Extension Audit Report	Scheduled evaluation	`rb6@duke.edu`	Complete roster audit detailing students eligible for semester-end extensions (until Friday, Dec 4, 2026 at 11:59 PM) based on > 2 quiz NPs or > 4 worksheet NPs. **Students are never emailed**; only Rann is notified.
Quiz Late Retake Setup Alert	Upon assignment creation	`rb6@duke.edu`	Confirms creation of `Quiz <N> late retake` with availability set for December 4–5, 2027 (1 year post-term) with old extensions cleared.
Status Report Sample Previews	Prior to broadcast	`rb6@duke.edu`	Dispatches representative sample student status emails to Rann for verification prior to full-class batch transmission.
Election & Switchover Infrastructure Alerts	Daily EV morning / Election Day	`rann@rannbar-on.com`	Status reports for civic web platforms: daily EV turnout model shifts, automated transition to Election Day polling locations, and API token refreshes.

5 Strict Governance Policies & System Guardrails

All automated scripts, background agents, and cron tasks operate under strict programmatic guardrails established in the global environment rules:

Rule 1: Non-Current Assignments Absolute Prohibition

NEVER send any notifications whatsoever regarding non-current assignments to any students in any course. Deadlines, extensions, submission window modifications, and student notifications **MUST ONLY** occur for **CURRENT** assignments unless explicitly instructed in writing by Rann. All extensions on non-current assignments must remain removed.

Rule 2: Permanently Dropped Students Policy

Isaiah Nwoko (`isaiah.nwoko@duke.edu`), Marissa Payne (`marissa.payne@duke.edu`), and Emmanuela Kumi (`emmanuela.kumi@duke.edu`) have been permanently removed as students from Math 105L. They must **NEVER** receive any notifications, extensions, retake opportunities, or status reports under any circumstances across all pipelines.

Rule 3: Semester-End Extensions Policy

Extensions until the end of the semester (Friday, December 4, 2026 at 11:59 PM) are granted **ONLY** to students who have either:

- More than two (> 2) no-passes on quizzes, OR
- More than four (> 4) no-passes on worksheets.

They are granted strictly for qualifying assignments and qualifying students. **In all such cases, NEVER email students.** Email Rann (`rb6@duke.edu`) with the full audit report.

Rule 4: Quiz Late Retake Architecture (No Direct Extensions for Quizzes)

For quizzes, **NEVER** extend original deadlines. Instead, duplicate the original quiz as `Quiz <N> late retake`. Configure its submission window from one year after classes end until the following day (December 4, 2027 at 11:59 PM to December 5, 2027 at 11:59 PM). Remove all extensions on the original quiz.

Rule 5: Quiz Retake Score Download Delay Rule

Quiz retake scores are not downloaded to `rawscores/` or released to students until **one week and one hour** after the retake assignment first opens. This delay is guarded programmatically in `gradescope_tasks.py` by `is_quiz_retake_eligible_for_download()`.

6 Summary of System File Artifacts

- **Master LaTeX Document:** `/home/rann/src/gradescopetools/notifications_overview.tex`
- **Compiled PDF Document:** `/home/rann/src/gradescopetools/notifications_overview.pdf`
- **Duke Website Publication:** `/home/rann/src/Dukewebsite/static/notifications_overview.pdf`
- **Main Website Publication:** `/home/rann/src/website/mainwebsite/notifications_overview.pdf`
- **Course Schedule Integration:** `/home/rann/src/Dukewebsite/content/project/105lmaterials.md`